

Bilal Ahmed

Robotics Software & Research Engineer

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Professional Summary

Robotics software and research engineer specializing in autonomous systems, multi-sensor localization, robot perception, mission planning, and embodied AI. Experience developing and evaluating ROS/ROS 2 systems for underwater, ground, aerial, legged, and humanoid robots, including deployment on real AUV platforms. Published researcher and robotics competition award recipient with additional experience in LLM/VLM-based autonomy, reinforcement learning, simulation, and 3D reconstruction.

Technical Skills

Languages: Python, C++, MATLAB, C/Arduino

Robotics: ROS/ROS 2, Behavior Trees, localization, sensor fusion, SLAM, navigation, motion planning, task-priority control, inverse kinematics

Perception: OpenCV, PCL, stereo vision, depth/disparity, RGB-D, LiDAR, sonar, point clouds, object detection, visual odometry

AI & Robot Learning: PyTorch, TensorFlow, reinforcement learning, LLM/VLM reasoning, RAG, FAISS, BM25, knowledge graphs, OpenVLA

Simulation & Tools: OpenGL, MuJoCo, PyBullet, Gazebo, Stonefish, Docker, Git, Linux, Blender, SolidWorks, Simulink

Engineering: Modular architecture, object-oriented design, unit/integration testing, debugging, benchmarking, documentation

Research & Engineering Experience

Visiting Scholar – MAESTRO: Agentic Recovery for Marine Robotics

Feb. 2026 – Present

Ocean Systems Lab, Heriot-Watt University

Edinburgh, United Kingdom

- Developing an LLM/VLM-based multi-agent ROS 2 framework that detects mission anomalies, diagnoses faults, and generates recovery actions for autonomous underwater vehicles.
- Built a self-improving RAG-based ROS 2 code-generation and validation pipeline using DeepSeek-Coder-V2 and FAISS, achieving an 80% validated success rate across 50 generated nodes.
- Achieved 89% recovery-decision accuracy across 255 evaluation scenarios and improved the Coordinated Success Score from 2.42 to 3.92 through iterative multi-agent refinement.
- Integrated robot state, mission context, perception inputs, and knowledge-graph evidence for explainable diagnosis; benchmarked multiple models for diagnostic efficiency.

Research Intern – 3D Reconstruction & Multimodal Sensing

Aug. 2025 – Oct. 2025

COE MARBLE Lab, University of Zagreb

Zagreb, Croatia

- Developed a multimodal 3D reconstruction pipeline for offshore structures using stereo cameras, forward-looking sonar, and AUV pose estimates.
- Integrated FoundationStereo disparity with robot pose to produce dense georeferenced point clouds.
- Applied pose-aware particle filtering to reduce vertical ambiguity in sonar measurements and improve reconstruction consistency.
- Presented the work at Breaking the Surface 2025 after winning the RAMI research poster competition.

Volunteer Researcher – MiniGirona AUV

Nov. 2024 – Jul. 2025

CIRS Lab, University of Girona

Girona, Spain

- Contributed to localization, navigation, perception, mission planning, inspection, and manipulation on the MiniGirona autonomous underwater vehicle.
- Implemented EKF-based localization fusing DVL, INS, and mechanical imaging sonar point-to-line measurements in GPS-denied environments.
- Developed voting-based pose initialization and integrated path planning, task sequencing, and Alpha 5 manipulation behaviors for RAMI missions.

Education

Erasmus Mundus Joint Master – Intelligent Field Robotic Systems

Sep. 2024 – Present

University of Zagreb / University of Girona | GPA: 9.40/10.0

Croatia / Spain

Bachelor of Mechatronics Engineering

Oct. 2018 – Oct. 2022

Mehran University of Engineering and Technology | GPA: 3.9/4.0 | 2nd in Faculty

Jamshoro, Pakistan

Robotics Projects

Underwater & Marine Robotics

AUV Navigation with DVL, INS & Mechanical Imaging Sonar | *EKF, ROS, Sensor Fusion* Feb. 2025 – Jul. 2025

- Implemented EKF-based localization using DVL, INS, and sonar point-to-line updates for GPS-denied AUV navigation.
- Developed voting-based pose initialization and validated the pipeline in simulation and on the MiniGirona AUV.

Multimodal 3D Reconstruction of Offshore Structures | *Stereo Vision, Sonar, Particle Filtering* Aug. 2025 – Oct. 2025

- Fused FoundationStereo depth, forward-looking sonar, and AUV pose to reconstruct offshore structures as dense 3D point clouds.
- Used particle filtering to estimate sonar elevation and reduce vertical ambiguity in the reconstructed geometry.

Stereo Keypoint Extraction for Underwater Manipulation | *YOLOE, USLN, Stereo Disparity, PCL* Feb. 2025 – Jul. 2025

- Built an end-to-end pipeline for underwater enhancement, zero-shot detection, stereo depth, and 3D keypoint extraction for valves, buoys, and damaged structures.
- Estimated task-relevant 3D features using clustering, PCA, stereo geometry, and point-cloud processing.

Color Enhancement for Underwater Robot Perception | *OpenCV, HSV, RGB, YCbCr* Feb. 2025 – Jul. 2025

- Implemented and compared color-space enhancement methods to improve underwater contrast and downstream detection reliability.

Robot Simulation & Autonomous Navigation

MarSim – Mars Robotics Simulator | *OpenGL, ROS 2, Physics, Sensor Simulation* 2025 – Present

- Developed a lightweight Mars robotics simulator with custom terrain rendering, rover dynamics, environmental effects, and ROS 2 interfaces.
- Implemented virtual RGB-D and stereo cameras, LiDAR, IMU, joint states, and ground-truth odometry for perception and navigation experiments.
- Built reusable interfaces for deploying localization, mapping, planning, and control algorithms without relying on Gazebo, MuJoCo, or Isaac Sim.

Uncertainty-Aware Active Robot Navigation | *ROS 2, Active Perception, Safety-Aware Planning* May 2026

- Developed a ROS 2 navigation system that estimates uncertainty from depth and LiDAR observations and incorporates it into path selection.
- Implemented active viewpoint selection before entering ambiguous or occluded regions and evaluated collision rate, success, efficiency, and uncertainty.

Memory-Augmented Human-Aware Navigation | *Adaptive Planning, Intent Prediction* Jul. 2026

- Built a partially observable navigation benchmark combining reactive avoidance with long-horizon temporal and semantic reasoning.
- Reduced emergency interventions by 92% and eliminated occluded surprises compared with a reactive baseline, with a 2.2% increase in completion time.

LLM/VLM-Based Autonomy & Human–Robot Interaction

Can Robots Code? – Self-Improving ROS 2 Coding Agent | *LLM, RAG, Validation* Feb. 2026 – Jul. 2026

- Built a self-improving pipeline that converts brief prompts into detailed ROS 2 node specifications and executable code.
- Evaluated 50 tasks: 50/50 prompts expanded correctly, 45/50 nodes were syntactically valid, and 40/50 executed successfully.

TiAGo LLM–VLM Smart-Home Assistant | *ROS, LLM, VLM, HRI* Sep. 2025 – Feb. 2026

- Integrated language and vision models with face-activated and gesture-based interaction for task execution on the TiAGo platform.

CORTEX-R Warehouse Mission Executive | *Qwen, LangGraph, PyBullet, A** May 2026

- Developed a multimodal agentic executive that converts natural-language warehouse requests into verified multi-step robot missions.
- Integrated A* navigation, docking, lifting, battery-aware planning, worker-safety monitoring, approval handling, and recovery.
- Validated nine visual scenarios involving blocked aisles, moved inventory, low battery, damaged packages, worker crossings, and perception loss.

Robot Learning, Manipulation & Legged Robotics

OpenVLA-Assisted Pick-and-Place | *OpenVLA, PyBullet, KUKA* 2026

- Integrated RGB observations and language commands with OpenVLA to generate 7D actions for KUKA pick-and-place tasks.
- Implemented an inverse-kinematics adapter, multi-camera retry logic, and object-to-target success evaluation.

Humanoid Locomotion via Reinforcement Learning | *MuJoCo, PPO* 2026

- Trained humanoid policies for walking, goal reaching, and obstacle avoidance using a staged PPO curriculum.
- Progressed from stable locomotion to goal-directed navigation and collision-free obstacle avoidance through three training stages.

Unitree Go2 SLAM & RL-Based Navigation | *LiDAR, ROS 2, Legged Robotics* 2026 – Present

- Implemented LiDAR-based SLAM and autonomous indoor navigation for the Unitree Go2 quadruped.
- Exploring reinforcement-learning policies for obstacle avoidance during legged locomotion.

Task-Priority Underwater Manipulation | *Alpha 5 Arm, Inverse Kinematics*

2025

- Implemented task-priority control behaviors for valve and ring manipulation using the Alpha 5 arm on MiniGirona.

SLAM, Localization & Motion Planning

Stereo Visual Odometry & SLAM | *KITTI, Bundle Adjustment, Loop Closure*

Sep. 2025 – Feb. 2026

- Built a stereo odometry pipeline with feature tracking, stereo matching, triangulation, PnP, bundle adjustment, loop closure, and pose-graph optimization.
- Evaluated trajectory accuracy on KITTI using Absolute Trajectory Error and Relative Pose Error.

LiDAR SLAM on TurtleBot | *ROS 2, LiDAR*

Feb. 2025 – Mar. 2025

- Implemented LiDAR-based mapping and localization in simulation and extended the pipeline toward real TurtleBot deployment.

Comparative Robot Path Planning | *A*, RRT, RRT*, Wavefront, Potential Fields*

Sep. 2024 – Dec. 2024

- Implemented and compared five planners and integrated RRT/RRT* with a Behavior Tree for structured TurtleBot navigation.

Differential-Drive Localization | *PF, KF, EKF*

Sep. 2024 – Jan. 2025

- Compared particle, Kalman, and extended Kalman filters under noisy odometry in GPS-denied indoor environments.

Autonomous Navigation with Q-Learning | *Reinforcement Learning*

Sep. 2024 – Dec. 2024

- Implemented Q-learning for navigation in simulation and compared the learned policy with classical planning methods.

Multi-Agent, Swarm & Aerial Robotics

Crazyflie Consensus & Auction-Based Task Allocation | *Multi-Agent Systems, Drones*

Sep. 2025 – Feb. 2026

- Implemented consensus-based formation control and auction-based mission-level task allocation for a Crazyflie swarm.

Swarm Control with Reynolds Rules | *Separation, Alignment, Cohesion*

Sep. 2025 – Feb. 2026

- Implemented decentralized flocking behavior using separation, alignment, and cohesion in multi-agent simulation.

RL-Based PID Tuning for Drone Control | *Reinforcement Learning, PID*

Sep. 2025 – Feb. 2026

- Applied reinforcement learning to tune PID gains automatically for simulated drone altitude and position control.

Selected Undergraduate Engineering Projects

Integrated 3D Scanner & Printer | *LiDAR, CNC, CAD*

Nov. 2021 – Nov. 2022

- Led a four-person team in building a platform for scanning and reproducing objects without an existing CAD model, achieving 70% geometric accuracy.

Hexapod Robot for Mine Inspection | *Raspberry Pi, OpenCV, Sensors*

Jun. 2022 – Jul. 2022

- Built a six-legged surveillance robot with gas, temperature, ultrasonic, and camera sensing, controlled through RC and Wi-Fi.

Cartesian Manipulator with GUI & G-Code Control | *Python, CNC, Stepper Motors*

2021

- Built a three-axis Cartesian manipulator using stepper-driven prismatic joints and a Python interface for G-code execution.

Potato Disease Classification | *TensorFlow, CNN*

Oct. 2023 – Nov. 2023

- Trained a CNN with data augmentation to classify three potato-leaf disease classes with 90% accuracy.

Additional Mechatronics Projects | *Manipulators, CNC, IoT, Mobile Robots*

2018 – 2022

- Built 4-DOF and 6-DOF manipulators, a DVD-drive CNC machine, line-following and obstacle-avoidance robots, a human-following vehicle, a gesture-controlled car, and IoT systems.

Publications & Research Presentations

Advancing Accessible Underwater Robotics: The Mini-Girona I-AUV at RAMI 2025

Nov. 2025

ROBOT 2025 – 8th Iberian Robotics Conference

Porto, Portugal

- Peer-reviewed paper describing MiniGirona autonomous inspection, navigation, perception, and manipulation capabilities.

Multimodal 3D Reconstruction of Offshore Structures

Nov. 2025

Breaking the Surface Conference

Cyprus

- Invited and funded presentation following first place in the RAMI 2025 research poster competition.

Honours & Awards

RAMI Research Poster Competition – 1st Place <i>Research and Applications in Marine Robotics</i>	<i>Jun. 2025</i> <i>Cyprus</i>
RAMI Underwater Inspection & Maintenance Competition – 2nd Place <i>Autonomous Underwater Robotics Competition</i>	<i>Jun. 2025</i> <i>Cyprus</i>
Merit Certificate – 2nd Position in Mechatronics Faculty <i>Mehran University of Engineering and Technology</i>	<i>Dec. 2022</i> <i>Pakistan</i>

Hackathons & Technical Competitions

PlutusAudit AI – AI Agent Olympics <i>2nd Prize Milan AI Week</i> - Built a five-agent invoice-processing system using Gemini 2.5 Flash, Qwen3-32B, FAISS, BM25, and HyDE over 80+ accounting rules. - Implemented a SHA-256 hash-chained audit trail for tamper evidence and compliance-oriented traceability.	<i>May 2026</i> <i>Milan, Italy</i>
BobForBot – IBM Bob Hackathon <i>ROS 2 Code Analysis and Safety Automation</i> - Extended IBM Bob with four ROS 2 modes and a six-tool MCP server for inspecting binary robot mission files. - Identified and repaired an emergency-stop defect in a ROS 2 safety manager and generated automated documentation and safety checks.	<i>May 2026</i>
GeoFusion-MT – ESA Φ-lab GeoAI Challenge <i>Multitask Geospatial Learning</i> - Built a dual-branch fusion model with a UNet++ decoder for joint segmentation and height regression from foundation-model embeddings. - Added LightGBM residual correction using IncrementalPCA-compressed embeddings to improve height estimates.	<i>2026</i>